

Signal Sense – IoT Sensor Anomaly Detection Dashboard

Project & Technology Case Study

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Part 1 – Project Case Study

Problem Statement

IoT devices continuously generate sensor data such as temperature, humidity, pressure, voltage, vibration, and current. Detecting abnormal readings manually is difficult because of the huge volume of data. Organizations need an automated system that identifies abnormal sensor behavior quickly to prevent equipment failures and reduce maintenance costs.

Existing System Analysis

Many industries monitor sensor data using spreadsheets or basic dashboards. These systems display values but do not intelligently detect anomalies.

The existing systems generally:

- Display only raw sensor readings
- Require manual monitoring
- Lack predictive analysis
- Do not generate automatic alerts
- Provide limited visualization

Drawbacks of Existing System

- Manual monitoring consumes time.
- Human errors may occur.
- Equipment failure may remain undetected.
- Difficult to analyze thousands of sensor records.
- No machine learning support.
- Limited visualization.

Proposed Solution

Signal Sense is an AI-powered IoT Sensor Anomaly Detection Dashboard developed using Python, Flask, HTML, CSS, JavaScript, Pandas, Scikit-learn, and Plotly.

The system automatically:

- Reads IoT sensor data
- Cleans the dataset
- Detects anomalies using Machine Learning
- Displays interactive charts

- Generates alerts
- Shows normal and abnormal sensor values

Objectives of the Project

- Detect abnormal sensor readings automatically.
- Reduce manual monitoring.
- Improve equipment safety.
- Visualize sensor data.
- Generate anomaly reports.
- Provide a user-friendly dashboard.

Scope of the Project

The project can be used in:

- Smart Manufacturing
- Smart Agriculture
- Healthcare IoT
- Industrial Automation
- Smart Cities
- Power Plants
- Environmental Monitoring

Features and Modules

Module 1

Dataset Upload

Upload IoT sensor dataset.

Module 2

Data Preprocessing

- Remove missing values
- Clean data
- Normalize values

Module 3

Machine Learning

Detect anomalies using Isolation Forest.

Module 4

Visualization

Generate:

- Line Charts
- Bar Charts
- Scatter Plots
- Pie Charts

Module 5

Dashboard

Display:

- Sensor readings

- Detected anomalies
- Statistics
- Alerts

Workflow of the System

1. Upload Dataset



2. Data Cleaning



3. Feature Selection



4. Machine Learning Model



5. Anomaly Detection



6. Visualization



7. Dashboard Output

Expected Outputs

- Sensor Dashboard
- Cleaned Dataset
- Detected Anomalies

- Interactive Graphs
- Performance Report

Future Enhancements

- Real-time IoT Integration
- Email Alerts
- SMS Notifications
- Deep Learning Models
- Cloud Deployment
- Mobile Application
- Predictive Maintenance

Conclusion

Signal Sense provides an intelligent solution for monitoring IoT sensor data. By combining Flask with Machine Learning, the system detects anomalies efficiently, reducing manual work and improving equipment reliability.

Part 2 – Technology Case Study

Flask Framework

Introduction

Flask is a lightweight Python web framework used to develop web applications quickly.

Features

- Lightweight
- Easy Routing
- Jinja Templates
- REST API Support
- Debug Mode
- Secure Sessions

Advantages

- Easy to Learn
- Flexible
- Fast Development
- Large Community
- Python Integration

Why Flask is Suitable

Flask is suitable because it:

- Connects Machine Learning models
- Displays dashboards
- Reads datasets
- Generates web pages
- Supports interactive applications

HTML

Purpose

HTML creates the structure of web pages.

Important Tags

- `<html>`
- `<head>`
- `<body>`
- `<form>`
- `<table>`
- `<div>`
- `<input>`
- `<button>`
- `<nav>`
- `<img>`
- `<h1>`
- `<p>`

Form Elements

- Text Box
- File Upload
- Submit Button
- Dropdown
- Radio Button
- Checkbox

Tables

Used to display sensor data.

Cards

Used to display:

- Total Sensors
- Total Anomalies
- Alerts

Navigation

Navigation bar helps users move between dashboard pages.

CSS

Purpose

CSS improves the appearance of the website.

Responsive Design

Responsive design automatically adjusts layouts for:

- Mobile
- Tablet
- Laptop
- Desktop

Flexbox

Used for arranging dashboard cards.

Grid

Used for arranging charts.

UI/UX Principles

- Clean Layout
- Good Typography
- Proper Colors
- Easy Navigation
- User Friendly

Professional Practices

- Consistent Colors
- Responsive Layout
- Proper Spacing
- Attractive Dashboard
- Fast Loading

JavaScript

Purpose

JavaScript makes webpages interactive.

DOM Manipulation

Update webpage content dynamically.

Form Validation

Check uploaded files before processing.

Dynamic Content

Display charts and anomaly results without refreshing the page.

Event Handling

Handle:

- Button Click
- File Upload
- Mouse Events
- Keyboard Events

Flask Template Rendering

Project Structure

Signal Sense/

```
|
|— app.py
|— templates/
|   index.html
|   dashboard.html
|— static/
```

```
|       style.css
|       script.js
|----- dataset/
|       sensor_data.csv
|----- model/
```

Templates Folder

Stores HTML pages.

Static Folder

Stores:

- CSS
- JavaScript
- Images

render_template()

Used to display HTML pages.

Example:

```
return render_template("dashboard.html")
```

Jinja Templating

Example:

```
{{ sensor_value }}
```

Passing Data

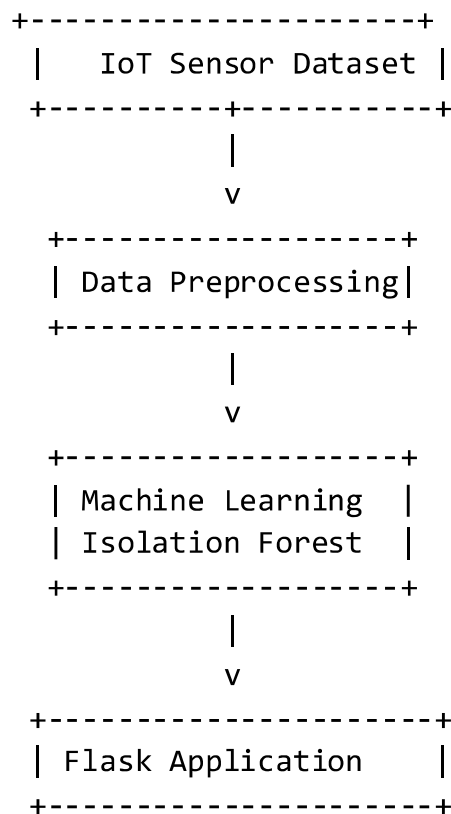
Example:

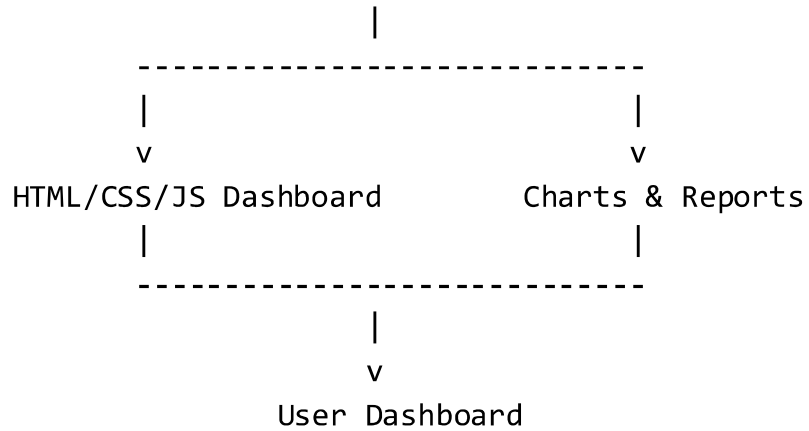
```
return render_template(  
    "dashboard.html",  
    data=sensor_data)
```

Dynamic Rendering

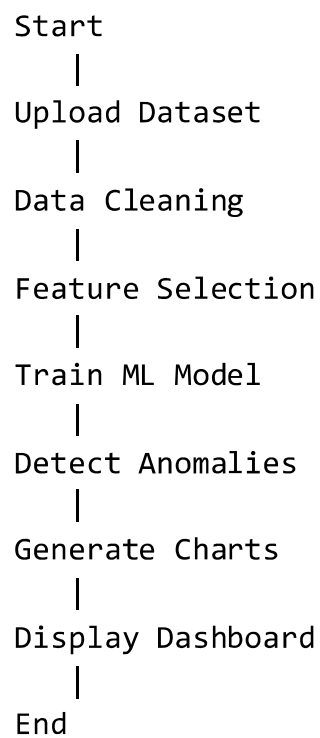
The webpage automatically updates using sensor values received from Flask.

Architecture Diagram





Workflow Diagram



Module Explanation

Module 1

Dataset Upload

Reads IoT sensor CSV file.

Module 2

Preprocessing

- Missing Values
- Data Cleaning
- Encoding

Module 3

Machine Learning

Isolation Forest detects abnormal records.

Module 4

Visualization

Displays interactive charts.

Module 5

Dashboard

Shows anomaly status and reports.

Reference Images



